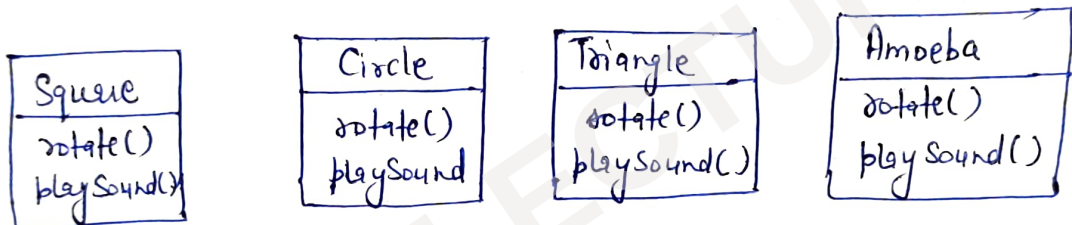


Inheritance in Java

- It allows one class to inherit the properties & behaviors (fields & methods) of another class.
- It helps in code reusability & establishes a parent-child relationship between classes

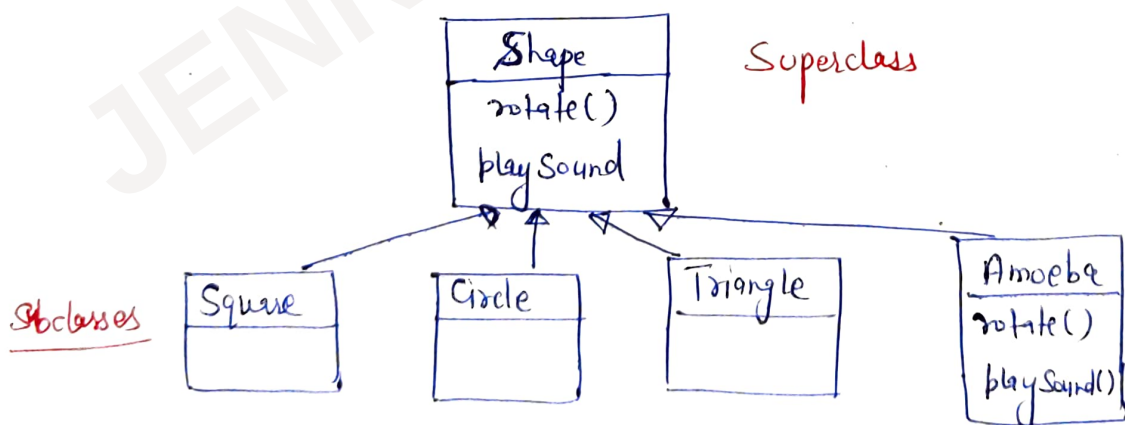
→ Let's revisit our example discussed in lecture 6 (High level Overview of OOP)

Two employees (Nancy & Rahul) were given a task and Rahul followed OOP approach to do that task.



But the problem here was :- duplicacy of code

then the updated version was :-



Here Amoeba class is overriding the rotate() & playSound() methods because amoeba rotates in a different way from other shapes

Here we can say "Square inherits from Shape"
 "Circle inherits from Shape" etc

Here Shape is Superclass or Parent class or Baseclass & all the four (Square, Circle, Triangle, Amoeba) are sub classes, child classes or derived classes. It means if the shape class has the functionality then the subclasses automatically get that same functionality.

Advantages:-

- ① Code Reusability
↳ eliminates redundancy
- ② Method Overriding
↳ child classes can provide specific implementation for methods in the parent class enabling runtime polymorphism
- ③ Improved Maintainability
↳ if you need to change then you change only in parent class & that would be reflected in its child class
- ④ Extensibility
↳ New features can be added to existing classes without modifying their source code
- ⑤ Code Organization
↳ It allows hierarchical classification of classes, making the program easier to understand & manage. It also helpful to model real world objects & relationships.

But if the inheritance hierarchy is deep then it can make the code complex & hard to understand.

Superclass / Parent class / Base class :-

- It is a class from which other classes inherit. It contains common properties & methods that can be shared with its subclasses.

eg:- Animal is a superclass for Dog & cat etc.

Subclass / Child class / Derived class :-

- It inherits from Superclass. It can reuse or override the methods ~~and fields~~ of the superclass & can also define additional properties & methods of its own.
- Instance variables are not overridden. They don't define any special behavior, so a subclass can give an inherited instance variable any value.

Types of Inheritance :-

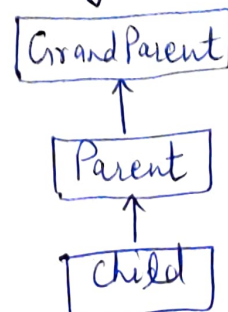
- ① Single Inheritance :- A child class inherits from a single parent class.

A $\Rightarrow$ Parent class

B $\Rightarrow$ child class

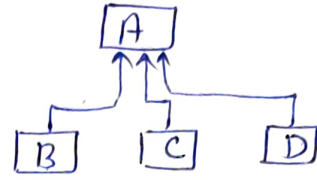


- ② Multilevel Inheritance :- A class inherits from a class & another class inherits from that derived class forming a chain



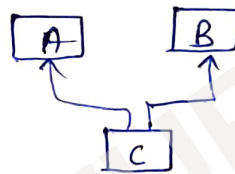
③ Hierarchical Inheritance: - Multiple child classes inherit from a single Parent class

A is Parent class &
B, C, D are derived classes.



NOTE: - Multiple Inheritance: - Java doesn't support multiple inheritance with classes. But we can achieve multiple inheritance through interfaces.

A, B are Parent classes &
C is child class



So in this one child class inherits from more than one parent classes. It can cause ambiguity (Diamond Problem)

Hybrid Inheritance: - Combination of more than one type of inheritance. But in Java this is possible only through interfaces.

IMP Inheritance represents IS-A relationship.

eg:- Student IS-A person

→ But if there is no IS-A relationship between two classes then it is not recommended to use inheritance.

Inheritance IS-A relationship works in only one direction e.g:-

Circle IS-A shape → make sense but reverse is not making any sense

Shape IS-A Circle → doesn't make sense.

So here Circle extends Shape but Shape should not extend Circle.

NOTE:- In Java Every class implicitly inherits from the object class, which is root of the class hierarchy.

The object class provides several important methods such as:
toString(), equals(Object obj), hashCode() etc.

JENNY'S LECTURES